Supplementary Tables

An Evaluation Artefact in Presence–Background Archaeological Modelling: Evidence from East Java and a Corrected Comparison Protocol

Every table below is generated directly from the raw result files by `build_supplement.py`, which is included in the code repository cited in the manuscript's Data Availability statement. No value is retyped. Tables S3–S6 describe the refutation suite (E217–E224); backgrounds there are drawn from a decimated raster lattice rather than by continuous rejection sampling, so absolute AUC levels are not directly comparable with the published E007–E013 pipeline — the interpretable quantities are the within-experiment contrasts. This is stated in the manuscript at §2.6.

Seed	hard_frac (realised)	XGB AUC	XGB TSS	RF AUC	RF TSS
375	0.619	0.768	0.507	0.742	0.458
5621	0.614	0.752	0.476	0.743	0.454
8971	0.624	0.752	0.478	0.762	0.494
10883	0.601	0.742	0.451	0.739	0.456
11903	0.604	0.742	0.438	0.730	0.438
17010	0.609	0.738	0.458	0.731	0.424
20828	0.615	0.735	0.435	0.742	0.457
21924	0.609	0.729	0.431	0.733	0.454
22830	0.623	0.774	0.487	0.747	0.470
25916	0.621	0.774	0.500	0.760	0.486
26138	0.610	0.755	0.458	0.734	0.446
27291	0.635	0.753	0.463	0.755	0.445
32581	0.617	0.753	0.472	0.748	0.466
33850	0.612	0.740	0.474	0.733	0.459
36226	0.613	0.767	0.486	0.759	0.464
39446	0.623	0.759	0.465	0.754	0.446
39691	0.615	0.749	0.460	0.742	0.461
41287	0.616	0.739	0.441	0.749	0.458
41432	0.611	0.743	0.459	0.745	0.469
44036	0.622	0.752	0.466	0.741	0.448

XGBoost AUC across the 20 seeds: min 0.729, max 0.774, mean 0.751. Realised hard-negative fraction is consistently near 0.62, well above the 0.30 target – see manuscript §2.3.

Table 1: Table S1. Seed stability of E013 (the top rung of the reported ladder). Each row is one background-generation seed; AUC and TSS are means over the five deterministic spatial-CV folds. Source: `supplement/e013_seed_stability.csv`.

Block size	deg	km	Runs	XGB AUC	XGB 95% CI	RF AUC	Δ XGB vs 50 km
40km	0.360	40.0	20	0.725	[0.718, 0.733]	0.742	−0.025
50km baseline	0.450	50.0	20	0.751	[0.746, 0.757]	0.744	+0.000
60km	0.541	60.0	20	0.742	[0.737, 0.747]	0.732	−0.009

Table 2: Table S2. Block-size sensitivity of E013 under spatial block cross-validation. The reported result uses the 50 km baseline; performance falls at both 40 km and 60 km, so the baseline is not a conservative choice. Source: `supplement/e013_blocksize_summary.csv`.

Feature set	Background design	Algorithm	AUC (own bg)	AUC (common bg)	Own − common
terrain	hybrid	maxent	0.6709	0.6324	+0.0384
terrain	hybrid	randomforest	0.7137	0.6833	+0.0303
terrain	hybrid	xgboost	0.7165	0.6830	+0.0335
terrain	random_buffered	maxent	0.6177	0.6234	−0.0057
terrain	random_buffered	randomforest	0.6716	0.6730	−0.0015
terrain	random_buffered	xgboost	0.6750	0.6783	−0.0034
terrain	random_nobuffer	maxent	0.6204	0.6284	−0.0081
terrain	random_nobuffer	randomforest	0.6672	0.6730	−0.0057
terrain	random_nobuffer	xgboost	0.6704	0.6813	−0.0109
terrain	tgb	maxent	0.6147	0.6227	−0.0080
terrain	tgb	randomforest	0.6696	0.6718	−0.0022
terrain	tgb	xgboost	0.6741	0.6759	−0.0017
terrain_river	hybrid	maxent	0.7050	0.6634	+0.0417
terrain_river	hybrid	randomforest	0.7403	0.7062	+0.0341
terrain_river	hybrid	xgboost	0.7497	0.7080	+0.0416
terrain_river	random_buffered	maxent	0.6893	0.6890	+0.0002
terrain_river	random_buffered	randomforest	0.7143	0.7152	−0.0009
terrain_river	random_buffered	xgboost	0.7120	0.7154	−0.0034
terrain_river	random_nobuffer	maxent	0.6905	0.6953	−0.0048
terrain_river	random_nobuffer	randomforest	0.7030	0.7094	−0.0064
terrain_river	random_nobuffer	xgboost	0.7066	0.7155	−0.0089
terrain_river	tgb	maxent	0.6865	0.6906	−0.0041
terrain_river	tgb	randomforest	0.7103	0.7136	−0.0033
terrain_river	tgb	xgboost	0.7129	0.7134	−0.0004

Own-background evaluation exceeds common-background evaluation in 7/24 cells (mean +0.0059). Backgrounds are drawn from a decimated (∼300 m) lattice, so absolute levels are not comparable to Table 3 of the manuscript; the interpretable quantities are the within-table contrasts (manuscript §2.6).

Table 3: Table S3. E217 common-background benchmark: every background design and feature set scored both on its own background and on a shared evaluation background, under identical spatial-block folds. This is the comparison the reviewer's Maximum Entropy request made possible. Source: `E217_maxent_benchmark/results/e217b_summary.csv`.

Algorithm	Trained on	Eval: uniform	Eval: tgb	Eval: hybrid	Eval: stratified
maxent	hybrid	0.6605	0.6626	0.7051	0.6750
maxent	random	0.6939	0.6891	0.6985	0.7071
maxent	tgb	0.6909	0.6867	0.6960	0.7058
randomforest	hybrid	0.7025	0.6992	0.7334	0.7095
randomforest	random	0.7118	0.7063	0.7236	0.7151
randomforest	tgb	0.7144	0.7086	0.7253	0.7180
xgboost	hybrid	0.7060	0.7052	0.7414	0.7147
xgboost	random	0.7145	0.7089	0.7259	0.7189
xgboost	tgb	0.7187	0.7124	0.7285	0.7207

Read down the columns, not across the rows. Within any single evaluation background, the hybrid-trained model does not lead; it leads only in the column that is its own background (hybrid). That column-versus-row asymmetry is the evaluation artefact.

Table 4: Table S4. E218 own- versus common-background matrix: three training designs (rows) each scored against four evaluation backgrounds (columns), mean over 20 seeds, identical folds throughout. Source: `E218_evaluation_artefact/results/e218_stageA_auc_matrix.csv`.

Background config	Algorithm	n runs	Reported AUC	True AUC	Reported − true	Map Jaccard
hybrid	maxent	80	0.8173	0.5057	+0.3116	0.0138
hybrid	randomforest	80	0.8175	0.6173	+0.2002	0.1221
hybrid	xgboost	80	0.8029	0.5898	+0.2131	0.0380
random	maxent	20	0.8526	0.8345	+0.0181	0.6464
random	randomforest	20	0.8486	0.8295	+0.0190	0.5972
random	xgboost	20	0.8394	0.8193	+0.0201	0.4969
tgb	maxent	20	0.8470	0.8345	+0.0125	0.6315
tgb	randomforest	20	0.8415	0.8293	+0.0122	0.5881
tgb	xgboost	20	0.8310	0.8184	+0.0125	0.4907

Across all 360 synthetic runs the reported value exceeds the truth in 343/360 cases (95.3%), median +0.187, minimum -0.031. The published ladder gain under examination was +0.092.

Table 5: Table S5. E222 synthetic worlds of known ground truth: reported AUC (scored on the design's own background) against true AUC (scored against the known intensity surface), averaged within background configuration and algorithm. Source: `E222_synthetic_ground_truth/results/e222_runs.csv`.

Exp.	Feature set	`road_dist` a feature?	Background designs	Evaluation background	Seeds	Algorithms
E217	terrain / terrain+river	no	random, tgb, hybrid	own + common	5	MaxEnt, XGB, RF
E217b	terrain / terrain+river	no	random, tgb, hybrid	own + common	5	MaxEnt, XGB, RF
E218 A	terrain+river	no	random, tgb, hybrid	uniform, tgb, hybrid, stratified	20	MaxEnt, XGB, RF
E218 B	terrain+river	no	random, tgb, hybrid	common (block-size sweep)	20	XGB
E218 C	terrain+river	no	dissimilarity sweep	common	5	XGB
E218 D	terrain+river	no	random, tgb, hybrid	own + common (∼150 m lattice)	5	XGB
E218b	terrain+river	no	hybrid, `hard_frac` 0.0–1.0	own + common	5	XGB
E220	terrain+river	no	`hard_frac` dial	own	5	XGB
E221	terrain+river	no	random, tgb, hybrid	own	10	MaxEnt, XGB, RF
E222	synthetic covariates	no	random, tgb, hybrid, quota	own + ground truth	30	MaxEnt, XGB, RF
E222c/d	synthetic covariates	no	random, tgb, hybrid, quota	own + ground truth	30	MaxEnt, XGB, RF
E223	terrain+river	no	random, tgb, hybrid	own + common (bootstrap)	30	MaxEnt, XGB, RF
E224	synthetic covariates	**yes** (the manipulation)	random, tgb	own + ground truth	30	MaxEnt, XGB, RF

`road_dist` is a training feature in E224 only, and there only because withholding it was the explanation under test; the test failed (manuscript §3.5). Everywhere else it builds the background and is never a predictor, exactly as in E007–E013.

Table 6: Table S6. Per-experiment design matrix for the refutation suite E217–E224, extending Table 2 of the manuscript. Built by reading the analysis scripts, not the prose.